In [ ]:

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import pandas as pd
from openpyxl import Workbook
from openpyxl.styles import Font, Alignment, PatternFill, Border, Side
from openpyxl.utils.dataframe import dataframe_to_rows
from openpyxl.utils import get_column_letter
from datetime import datetime
import glob
import warnings
import re
from copy import copy

warnings.filterwarnings('ignore')

# ==================== 配置区域 ====================
OUTPUT_PREFIX = "2-统计分析1"
REVIEWER = "ly"
CHECKER = "zy"

# 列名别名映射表（通用模板的核心）
COLUMN_ALIASES = {
    '户名': ['户名', '客户姓名', '账户名', '账户名称'],
    '卡号': ['卡号', '账号', '客户账号', '主卡卡号'],
    '日期': ['交易日期', '交易时间', '日期', '时间', '交易时刻'],
    '收入': ['收入', '转入', '贷方发生额', '存入', '存入金额', '收'],
    '支出': ['支出', '转出', '借方发生额', '支取', '支取金额', '付'],
    '余额': ['余额', '账户余额', '当前余额', '联机余额'],
    '金额': ['金额', '交易金额', '发生额', '交易本金'],
    '借贷标志': ['借贷标志', '借贷', '借/贷', '进出标志', '进出'],
    '对方户名': ['对方户名', '对方名称', '对方全称', '对方账号名称'],
    '对方账号': ['对方账号', '对方卡号', '对方账户'],
    '对方行名': ['对方行名', '对方行', '对方开户行', '对方银行名称', '对方机构'],
    '摘要': ['摘要', '交易备注', '文字摘要', '备注', '用途']
}

# ==================== 工具函数 ====================

def get_col(df, key):
    for alias in COLUMN_ALIASES.get(key, []):
        for col in df.columns:
            if alias == str(col).strip():
                return col
    return None

def parse_amount(val):
    if pd.isna(val) or val == '' or val is None: return 0
    try:
        if isinstance(val, str):
            val = val.replace(',', '').replace('¥', '').replace('$', '').replace
            if val.startswith('(') and val.endswith(')'): val = '-' + val[1:-1]
        return float(val)
    except: return 0

def parse_date(val):
    if pd.isna(val) or val == '' or val is None: return pd.NaT
    try:
        if isinstance(val, datetime): return val
        val_str = str(val).strip()
        if len(val_str) >= 8 and val_str[:8].isdigit():
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            if len(val_str) == 8: return pd.to_datetime(val_str, format='%Y%m%d'
            return pd.to_datetime(val_str[:14])
        return pd.to_datetime(val)
    except: return pd.NaT

def set_border(ws, range_str, style='thin'):
    side = Side(style=style)
    border = Border(left=side, right=side, top=side, bottom=side)
    for row in ws[range_str]:
        for cell in row: cell.border = border

def apply_standard_font(ws):
    for row in ws.iter_rows():
        for cell in row:
            is_bold = cell.font and cell.font.bold
            cell.font = Font(name='宋体', size=10, bold=is_bold)

def main():
    print("=" * 60)
    print("星语 - 司法鉴定流水专业格式化工具 (V8.0 布局精调版)")
    print("=" * 60)

    excel_files = [f for f in glob.glob("*.xlsx") + glob.glob("*.xls")
                   if not f.startswith('2-统计分析') and not f.startswith('5-统计
    if not excel_files: return print("错误: 未找到 Excel 源文件! ")

    if len(excel_files) == 1:
        source_file = excel_files[0]
        print(f"\n自动处理: {source_file}")
    else:
        print("\n请选择要处理的文件: ")
        for i, f in enumerate(excel_files, 1): print(f"  {i}. {f}")
        try:
            choice = int(input("\n输入编号: ")) - 1
            source_file = excel_files[choice]
        except (ValueError, IndexError):
            print("输入无效。")
            return

    df = pd.read_excel(source_file)
    c_name, c_card, c_date, c_inc, c_out, c_bal, c_amt, c_flag = [get_col(df, k)

    account_name = str(df[c_name].iloc[0]) if c_name else "未知"
    card_number = ''.join(filter(str.isdigit, str(df[c_card].iloc[0]))) if c_car
    bank_name = "中国民生银行" if "民生银行" in source_file else "未知开户行"

    n = len(df)
    new_df = pd.DataFrame()
    new_df['交易日期'] = df[c_date].apply(parse_date) if c_date else pd.NaT

    income, outcome = pd.Series([0.0]*n), pd.Series([0.0]*n)
    if c_inc and c_out:
        income, outcome = df[c_inc].apply(parse_amount), df[c_out].apply(parse_a
    elif c_amt and c_flag:
        raw_amt = df[c_amt].apply(parse_amount)
        for i in range(n):
            if str(df[c_flag].iloc[i]) in ['进', '贷', '收入']: income.iloc[i] =
            else: outcome.iloc[i] = raw_amt.iloc[i]
    elif c_amt:
        raw_amt = df[c_amt].apply(parse_amount)
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        income, outcome = raw_amt.apply(lambda x: x if x > 0 else 0), raw_amt.ap

    final_df = pd.DataFrame()
    final_df['户名'] = [account_name] * n
    final_df['账号'] = [card_number] * n
    final_df['币种'] = ['CNY'] * n
    final_df['交易日期'] = new_df['交易日期']
    final_df['发生额'] = income - outcome
    final_df['收入'] = income
    final_df['支出'] = outcome
    final_df['账户余额'] = df[c_bal].apply(parse_amount) if c_bal else 0
    final_df['对方户名'] = df[get_col(df, '对方户名')].fillna('') if get_col(df,
    final_df['对方账号'] = df[get_col(df, '对方账号')].fillna('') if get_col(df,
    final_df['对方开户行'] = df[get_col(df, '对方行名')].fillna('') if get_col(df
    final_df['交易摘要'] = df[get_col(df, '摘要')].fillna('') if get_col(df, '摘要
    final_df['文字摘要'] = ''
    final_df['余额勾稽'] = ''

    wb = Workbook()
    if wb.active: wb.remove(wb.active)

    # 样式定义
    thin = Side(style='thin')
    medium = Side(style='medium')
    blue_f = PatternFill(start_color='B7DEE8', end_color='B7DEE8', fill_type='sc
    yellow_f = PatternFill(start_color='FFFF00', end_color='FFFF00', fill_type='
    crimson_fill = PatternFill(start_color='F2DCDB', end_color='F2DCDB', fill_ty

    align_center = Alignment(horizontal='center', vertical='center')
    align_left = Alignment(horizontal='left', vertical='center')
    font_10 = Font(name='宋体', size=10)
    font_10_bold = Font(name='宋体', size=10, bold=True)
    font_title = Font(name='宋体', size=12, bold=False)

    sd = final_df['交易日期'].min().strftime('%Y/%m/%d') if not final_df['交易日其
    ed = final_df['交易日期'].max().strftime('%Y/%m/%d') if not final_df['交易日其
    today = datetime.now().strftime('%Y/%m/%d')
    out_name = f"{OUTPUT_PREFIX}-{account_name}-{datetime.now().strftime('%m%d')

    # ==================== 0-封面页 ====================
    ws = wb.create_sheet("0-封面")
    # 列宽设置
    ws.column_dimensions['A'].width = 15.4
    ws.column_dimensions['B'].width = 30.33
    ws.column_dimensions['C'].width = 23.44
    ws.column_dimensions['D'].width = 10.29
    ws.column_dimensions['E'].width = 12.73
    ws.column_dimensions['F'].width = 19.29
    ws.column_dimensions['G'].width = 14.94
    ws.column_dimensions['H'].width = 11.19
    ws.column_dimensions['I'].width = 14.74
    ws.column_dimensions['J'].width = 8.71

    # 行高与标题
    ws.row_dimensions[1].height = 13.1
    ws.merge_cells('A1:J1')
    ws['A1'] = '司法鉴定程序表'
    ws['A1'].font = font_title; ws['A1'].alignment = align_center

    # 核心矩阵 (Row 3-5)
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for r in [3, 4, 5]: ws.row_dimensions[r].height = 24.5
ws['A3'] = '所属会计师事务所: '; ws.merge_cells('B3:J3'); ws['B3'] = '立信会计
ws['A4'] = '被鉴定单位: '; ws.merge_cells('B4:C4'); ws['B4'] = '-'; ws['E4']
ws['A5'] = '鉴定账户: '; ws['B5'] = account_name; ws['C5'] = card_number; ws[
set_border(ws, 'A3:J5', 'medium')
for r in range(3, 6):
    for c in range(1, 11):
        if not ws.cell(r, c).alignment: ws.cell(r, c).alignment = align_cent

# 账户信息 (一、账户信息)
ws.row_dimensions[7].height = 13.1; ws['A7'] = '一、账户信息'; ws['A7'].font
for r in [8, 9]: ws.row_dimensions[r].height = 26
headers = ['资料名称', '账号', '开户银行', '流水起始日', '流水截止日', '期间要求
for i, h in enumerate(headers, 1):
    c = ws.cell(8, i, h); c.fill, c.border, c.alignment, c.font = blue_f, Bo
data_v = [account_name, card_number, bank_name, sd, ed, '√', '√', '', '√', '
for i, v in enumerate(data_v, 1):
    c = ws.cell(9, i, v); c.fill, c.border, c.alignment = yellow_f, Border(l

# 总体目标 (二、总体目标)
ws.row_dimensions[11].height = 13.1; ws['A11'] = '二、总体目标'; ws['A11'].fo
ws.row_dimensions[12].height = 61.05; ws.merge_cells('A12:J12'); set_border(

# 具体程序 (三、具体程序)
ws.row_dimensions[16].height = 13.1; ws['A16'] = '三、具体程序'; ws['A16'].fo
ws.row_dimensions[18].height = 13.1
p_headers = ['序号', '程序内容', '附表', '完成进度', '情况说明', '备注']
for i, h in enumerate(p_headers, 1):
    c = ws.cell(18, i, h); c.fill, c.border, c.alignment = blue_f, Border(le
for r in range(19, 24):
    ws.row_dimensions[r].height = 19.05
    ws.cell(r, 1, str(r-18)) # 1-5
    for c in range(1, 7): ws.cell(r, c).border = Border(left=thin, right=thi
ws.cell(19, 2, "账户余额勾稽查验"); ws.cell(19, 3, "1-统计分析")

# 结论 (四、鉴定结论)
ws.row_dimensions[26].height = 13.1; ws['A26'] = '四、鉴定结论'; ws['A26'].fo
ws.row_dimensions[28].height = 50; ws.merge_cells('A28:J28'); set_border(ws,
apply_standard_font(ws)

# =================== 1-统计分析页 ===================
ws2 = wb.create_sheet("1-统计分析")
# 列宽
ws2.column_dimensions['A'].width = 8.39
ws2.column_dimensions['B'].width = 17.07
for c in "CDE": ws2.column_dimensions[c].width = 8.39
for c in "FGHI": ws2.column_dimensions[c].width = 14.42
for c in "JK": ws2.column_dimensions[c].width = 8.39
ws2.column_dimensions['L'].width = 21.11
for i in range(13, 31): ws2.column_dimensions[get_column_letter(i)].width =

# 行高设置
ws2.row_dimensions[1].height = 28
for r in [2, 6]: ws2.row_dimensions[r].height = 11.25 # 移除第8行逻辑，第7行玑
ws2.row_dimensions[7].height = 22 # 表头稍微加高
for r in [3, 4, 5]: ws2.row_dimensions[r].height = 20

# 1. 标题 (Row 1)
ws2.merge_cells('A1:L1')
ws2['A1'] = '司法鉴定程序表'; ws2['A1'].font = font_title; ws2['A1'].alignmen
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    # 2. 核心矩阵 (Row 3-5) - 基于图片微调
    # Row 3
    ws2['A3'] = '所属会计师事务所：'
    ws2.merge_cells('B3:L3')
    ws2['B3'] = '立信会计师事务所（特殊普通合伙）深圳分所'

    # Row 4 (基于图片: A:标签, B-E:内容, F:标签, G:内容, H:标签, I:内容, K:标签, L:
    ws2['A4'] = '被鉴定单位：'
    ws2.merge_cells('B4:E4'); ws2['B4'] = '-'
    ws2['F4'] = '审核员：'; ws2['G4'] = REVIEWER
    ws2['H4'] = '日期：'; ws2['I4'] = today
    ws2['K4'] = '索引号：'; ws2['L4'] = ''

    # Row 5 (基于图片: A:标签, B-D:内容, E:标签, F-G:内容, H:标签, I:内容, J:标签,
    ws2['A5'] = '鉴定账户：'
    ws2.merge_cells('B5:D5'); ws2['B5'] = f"{account_name}{card_number}"
    ws2['E5'] = '鉴定期间'
    ws2.merge_cells('F5:G5'); ws2['F5'] = '-'
    ws2['H5'] = '复核员：'; ws2['I5'] = CHECKER
    ws2['J5'] = '日期：'; ws2['K5'] = today
    ws2['L5'] = '页次：'

    # 应用矩阵边框 (A3:L5)
    for r in range(3, 6):
        for c in range(1, 13):
            cell = ws2.cell(r, c)
            cell.border = Border(left=thin, right=thin, top=thin, bottom=thin)

    # 3. 数据表头 (Row 7)
    for i, h in enumerate(final_df.columns, 1):
        c = ws2.cell(7, i, h); c.font, c.fill, c.border, c.alignment = font_10_b
    ws2['P7'] = '期初余额'; ws2['P7'].fill = crimson_fill
    ws2['S7'] = '#REF!'; ws2['S7'].fill = crimson_fill

    # 4. 数据填充 (Row 8+) - 修正：第八行直接开始，行高上提
    for r_idx, row in enumerate(dataframe_to_rows(final_df, index=False, header=
        ws2.row_dimensions[r_idx].height = 15
        for c_idx, val in enumerate(row, 1):
            cell = ws2.cell(r_idx, c_idx, val)
            cell.font, cell.border = font_10, Border(left=thin, right=thin, top=
            if isinstance(val, (int, float)): cell.number_format = '#,##0.00'
            if c_idx == 14: # 余额勾稽 (N列)
                if r_idx == 8: cell.value = f"=H8-E8"
                else: cell.value = f"=H{r_idx}-E{r_idx}-H{r_idx-1}"
                cell.font = Font(name='Arial', size=9, color='FF0000')

    # 5. 全表居中对齐处理 (除非 B5/B3 特殊情况)
    for row in ws2.iter_rows():
        for cell in row:
            # 只有 B3, B5 可能需要特殊的左对齐（根据需求和美观）
            # 但用户最新指令是 "Sheet2 全表垂直和水平居中对齐"
            cell.alignment = align_center

    apply_standard_font(ws2)
    wb.save(out_name)
    print(f"\n✓ 处理完毕！输出文件：{out_name}")

if __name__ == "__main__": main()
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============================================================
星语 - 司法鉴定流水专业格式化工具 (V8.0 布局精调版)
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请选择要处理的文件:
  1. A.xlsx
  2. 6.xlsx
  3. 工作簿20.xlsx
  4. 新工作簿.xlsx
  5. 工作簿3.xlsx
  6. 深圳市嘉信达城投资发展有限公司交易明细.xls
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In [ ]: